Nested class:

- A nested class is a class which is declared in another enclosing class.
- A nested class is a member and as such has the same access rights as any other member.
- The members of an enclosing class have no special access to members of a nested class; the usual access rules shall be obeyed.

Syntax:

```
class OuterClass {
// Members of the outer class

public:
// Nested class definition

class InnerClass {
// Members of the inner class
};

};
```

Accessing Nested Classes

OuterClassName::NestedClassName objectName;

```
class Enclosing {
    class Nested { };
};
```

Example:

```
#include<iostream>
using namespace std;
class outer{
    public:
    int data=5;
    void display(){
        cout<<"I am outer class: "<<data<<endl;
    }
    class inner{
        public:
        void display(outer o){
int main(){
    outer o;
    o.display();
    outer::inner i;
    i.display(o);
    return 0;
}
```

```
#include<iostream>
using namespace std;
class A {
 public:
 class B {
private:int num;
public:
void getdata(int n) {
 num = n;
}
void putdata() {
 cout<<"The number"<<num;
}
```

```
int main() {
 cout<<"Nested classes in C++"<< endl;
 A :: B obj;
 obj.getdata(9);
 obj.putdata();
 return 0;
}
```

Output

Nested classes in C++

The number is 9

In the program, we will discuss how a member class can access class members(private) of the enclosing class.

```
#include <iostream>
using namespace std;
 //Declaring enclosing class
class Enclosing
{
private:
  int b;

  // Declaring Nested class
  class Nested
  {
  int y;
  void nestedAccess(Enclosing *a)
   {
     //private member of enclosed class can be accessed in nested class without any problem
     cout << "Nested Class" << a->b;
   }
  };
};

int main()
{
}
```

In the program, we will discuss how an enclosing class cannot access the private members of the nested class.

```
#include <iostream>
using namespace std;
//Declaring Enclosing class
class Enclosing{
private:
  int e;
  // Declaring Nested class
  class Nested  {
    int y;
  };
 void enclosingAccess(Nested *n)
   {
     //Member of nested class cannot be
accessed in enclosing class
     cout << "Nested Class" << n->y;
   }
};
int main()
{
}
```

Output:

```
error: 'int Enclosing::Nested::y' is private
within this context
      cout << "Nested Class" << n->y;
^
test.cpp:13:9: note: declared private here
    int y;
```

Member functions that are declared inside the nested function can be accessed outside the class using the scope resolution operator(::).

```
#include <iostream>
using namespace std;
//Declaring enclosing class
class Enclosing{
private:  int b;
  // Declaring Nested class
  class Nested  {
     public:     //Declaring member function inside but not defining it.
        int read();
        int good();    };    };
```

```
//Defining member functions outside the class
int Enclosing::Nested::read()
{
  cout<<"Read";
}
int Enclosing::Nested::good()
{
  cout<<"Good";
}

int main()
{
}
```

How to Define Nested Class as Private Member of Enclosing Class

```
#include <iostream>
using namespace std;
class Enclosing  {
private:
  class Nested   {
  public:
    void print()    {
      cout << "Nested class as private member" << endl;
    }   };
  Nested n ;
public:
  void show()
  {
    n.print();
  }
};
int main()
{
  Enclosing enc;
  enc.show();
  return 0;
}
```

Output:

```
Nested class as private member
```

How to Define Nested Class as Public Member of Enclosing Class?

```
#include <iostream>
using namespace std;
class Enclosing    {
//defined as public member of enclosing class
public:
  class Nested {
  public:
    void print() {
      cout << "Nested class as public member" << endl;
    }    };
};
int main()
{
  Enclosing::Nested en;
  en.print(); //calling member function of nested class
  return 0;
}
```

Output:

```
Nested class as public member
```

How to Define Member Functions of Nested Class Outside the Enclosing Class?

```
#include <iostream>
using namespace std;
class Enclosing   {
public:
  class Nested    {
  public:
    void print();
  };      };
//Member function defined outside the scope of class
void Enclosing::Nested::print()    {
  cout << "Nested class member outside " << endl;
  }
```

```
int main()
{
  Enclosing::Nested en;
  en.print(); //calling member function of nested cl
  return 0;
}
```

How to Define Nested Class Outside the Enclosing Class?

We can define the whole nested class outside the enclosing class with the help of the scope resolution operator.

```
#include <iostream>
using namespace std;
class Enclosing  {
public:
  class Nested;
};
class Enclosing::Nested     {
Public:     void print()   {
    cout << "Nested class outside enclosed class" << endl;       }      };
int main()
{
  Enclosing::Nested en;
  en.print(); //calling member function of nested class
  return 0;
}
```

Output:

```
Nested class outside enclosed class
```

Types of nested class:

Member Nested Classes

1) **Private Nested Class:** This is defined within the private section of the outer class.

2) **Public Nested Class:** This is defined within the public section of the outer class.

3) **Protected Nested Class:** This is defined within the protected section of the outer class.

Static Nested Classes

It's a class that is declared with the static keyword within an outer class. It does not have this pointer, therefore it cannot access non-static members of the outer class.

Local Nested Classes

These are classes that are defined inside a function or method, therefore they are local to a function.

Anonymous Nested Classes

These are unnamed classes, which are often used as an instance of a class with minimal implementation.

Friend-Nested Classes

It's a nested class that is declared as a friend within the outer class, which allows it to access private and protected members of the outer class.